ZCAS UNIVERSITY

MASTER OF SCIENCE IN COMPUTER SCIENCE

CIT 5612

DIGITAL FORENSICS AND CYBER CRIME INVESTIGATION

FINAL EXAMINATION

MONDAY 25TH NOVEMBER 2024

TIME: 14.00 – 17.00 HOURS

TIME ALLOWED: THREE HOURS (plus 5 minutes to read through the paper)

INSTRUCTIONS

1. Section A: Question One in Section A is **compulsory**.
2. Sections B: Answer Two (2) questions from this section.
3. This examination paper carries a total of **100 marks**.
4. Candidates must **not turn this page** until the invigilator tells them to do so.

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SECTION A: QUESTION 1 IS COMPULSORY AND MUST BE ATTEMPTED

Question 1:

Scenario: Read the scenario in Appendix I and then answer the following questions:

a) Explain how the Windows Registry can be utilized to identify evidence of persistence mechanisms used by malware. (6 Marks)

b) Discuss the registry hives you would analyze to understand a user's recent activities on a Windows system and why? (6 Marks)

c) Explain how you would use built-in Windows CLI commands to detect signs of privilege escalation? (6 Marks)

d) Discuss the benefits of using EnCase for acquiring digital evidence compared to traditional copying methods? (8 Marks)

e) Describe how EnCase facilitates keyword searches during a forensic investigation? (4 Marks)

f) Discuss the challenges that a forensic investigator faces when analyzing FAT file systems compared to NTFS? (10 Marks)

g) Explain how NTFS timestamps can be used to reconstruct a sequence of events on a compromised system. (10 Marks)

[Total : 50 Marks]

SECTION B: ATTEMPT ANY TWO (2) QUESTIONS FROM THIS SECTION

Question 2:

Refer to the scenario below and answer the questions that follow:

Scenario: Investigating a Corporate Espionage Case Using Mobile Device Forensics and IoT Artifacts

TechGlobal Inc., a leading tech firm, reported a data breach involving the unauthorized access and exfiltration of proprietary product blueprints. The company's internal investigation team discovered that sensitive data was transmitted from an employee's mobile device. Additionally, data logs from the office's IoT-enabled security cameras and smart access systems provided further evidence of suspicious activity during off-hours. The digital forensics team was brought in to investigate.

The forensic analysis involved extracting data from the suspected mobile device, including text messages, emails, app usage, and GPS location history. Additionally, investigators examined the security camera logs and smart door lock records, which revealed that the employee's mobile device had been used to access restricted areas of the building. The team also examined communication logs from the device, which included connections to various IoT devices. By piecing together evidence from the mobile device and IoT artifacts, the team aimed to establish the timeline and method of data exfiltration, as well as the extent of the breach.

a) Discuss the key data types that can be extracted from a mobile device in a forensic investigation? (10 Marks)

b) Explain the challenges of extracting data from a locked or encrypted mobile device. (10 Marks)

c) Describe how GPS location data from a mobile device can be utilized in an investigation like the one described in the scenario? (5 Marks)

[Total: 25 Marks]

Question 3:

Refer to the scenario below and answer the questions that follow:

Scenario: Analyzing a Network Breach at CyberBank Inc.

CyberBank Inc., a financial institution, experienced a sophisticated cyber attack in which sensitive customer information was exfiltrated from the corporate network. The attack was discovered when the bank's intrusion detection system (IDS) flagged a series of unusual outgoing data transfers during non-working hours. The digital forensics team was brought in to investigate the breach, focusing on network traffic analysis and application forensics.

Using packet capture tools and log analysis, the team discovered that the attacker had gained access through a vulnerable web application. The forensic analysis revealed a SQL injection attack that was used to bypass authentication mechanisms and execute malicious queries. Network logs showed that the attacker had established a reverse shell, enabling them to remotely control the compromised server and exfiltrate data. Furthermore, logs from the web server provided evidence of malicious scripts and unauthorized administrative actions. The forensic team had to piece together the attack sequence, identify the methods used, and determine the extent of the data breach.

a) Discuss the initial steps that a forensic investigator should take when analyzing a network intrusion like the one described? (10 Marks)

b) Explain the forensic evidence that would be critical in proving that a SQL injection attack occurred? (10 Marks)

c) Describe the artifacts you would analyze to determine the timeline of the attack? (5 Marks)

[Total: 25 Marks]

END OF EXAMINATION

Question 4:

Refer to the scenario below and answer the questions that follow:

Scenario: Digital Forensics & Cyber Crime Investigation

You are a digital forensics investigator tasked with analyzing the compromised system of a large financial institution. The breach occurred after an employee's computer was infected with a sophisticated piece of malware, which allowed the attacker to gain unauthorized access to the internal network. Upon further investigation, you discover various logs and event records related to the incident, as well as several password-protected files that may contain sensitive client information.

You begin your investigation by analyzing the system logs, which include login records, application events, network traffic, and suspicious activity that could indicate unauthorized access. You notice several failed login attempts followed by successful logins from unusual IP addresses, indicating that the attacker may have used a brute-force technique to crack the password. You suspect the attacker used a common password, and you need to employ password-cracking techniques to recover the password and gain full access to the system.

As part of your task, you must analyze the event logs to identify critical timestamps that show when the breach occurred, track the source of the attack, and determine how the malware was introduced into the system. You also need to perform password-cracking techniques to break into encrypted files that may contain evidence of financial fraud.

a) Describe the main password-cracking techniques and explain which one would be most effective in this scenario? (10 Marks)

b) After cracking a password and gaining access to sensitive files, discuss the steps you would take to ensure the evidence is preserved and handled properly? (10 Marls)

c) Discuss how you would use hash comparison to verify the integrity of password-protected files in this investigation? (5 Marks)

[Total: 25 Marks]

APPENDIX 1

Case Study: The Data Heist at FinSecure Corp.

FinSecure Corp., a prominent financial services company, experienced a severe data breach involving the exfiltration of client financial records. The attack was detected when an IT administrator noticed unusually high network activity on one of the main file servers. The internal security team was immediately called in to initiate a forensic investigation, focusing on critical areas: the Windows™ Registry, command-line interface (CLI) usage, digital evidence management using EnCase, and differences between FAT and NTFS file systems.

Windows™ Registry Analysis

The forensic team began by analyzing the Windows Registry on the compromised server. They focused on identifying modifications indicative of unauthorized backdoors or persistence mechanisms. The SYSTEM and SOFTWARE hives provided clues about installed software and services running at startup. Additionally, the analysis revealed evidence of changes to UserAssist keys, suggesting that the intruder used specific software to carry out the attack.

Working with Windows and CLI Systems

Upon reviewing CLI activity logs, the team discovered that the attacker had used advanced techniques to cover their tracks, such as disabling security features via the CLI and using obfuscated PowerShell commands. The analysis also showed that the attacker attempted to escalate privileges and moved laterally to other systems in the network using a combination of built-in Windows utilities and scripts.

EnCase Forensic Tools

The team used EnCase to create forensic images of the compromised systems. This allowed them to analyze evidence without risk of altering original data. Using EnCase's timeline feature, investigators reconstructed the sequence of events, pinpointing when the breach occurred and

tracking the attacker's movements. EnCase also helped identify hidden or encrypted files and recover deleted documents, which contained crucial evidence.

FAT and NTFS File System Examination

The attacker had used various removable storage devices to exfiltrate data, some of which were formatted using FAT32, while the main file servers ran on NTFS. An in-depth analysis of the NTFS file system provided metadata, such as MFT records, that helped establish access patterns and detect suspicious modifications. Conversely, examining FAT file systems proved more challenging, as the lack of robust security features made it difficult to ascertain detailed access information.

END OF CASE STUDY